Understanding Features

Data Features.....

- ‘titanic’ dataset in seaborn

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Learning Objectives

By the end of this session, you will be able to:

- Define categorical, discrete, and continuous features.
- Understand the differences and significance of each type.
- Explore examples with real-world datasets.
- Use Python (Pandas, NumPy, Matplotlib, Seaborn) for data exploration.

Types of Features

1

1. Categorical Features: Represent labels or categories (e.g., gender, country).

2

2. Discrete Features: Represent countable values (e.g., number of children).

3

3. Continuous Features: Represent measurable values (e.g., height, salary).

Categorical Features

- Definition: Features that represent categories or groups.

Examples:

- Gender: Male, Female
- Car Type: Sedan, SUV, Truck
- Payment Method: Credit Card, Cash, Digital Wallet

Data Features.....

- Which of these are Categorical features?

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
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deck	Categorical	Nominal	Represents the deck letter (A, B, C, etc.). While there might be a physical order, the deck itself is often treated as a set of distinct, unordered groups in analysis unless more context is given.
embark_town	Categorical	Nominal	Full name of the port of embarkation (Southampton, Cherbourg). No inherent order.
embarked	Categorical	Nominal	Represents the port of embarkation (S, C, Q). The ports have no inherent order.
sex	Categorical	Nominal	Represents distinct, unordered groups (male, female).
who	Categorical	Nominal	Represents a category (man, woman, child).
adult_male	Categorical	Nominal (or Binary/Dichotomous)	A logical category (True/False) indicating if the person is an adult male.
alive	Categorical	Nominal (or Binary/Dichotomous)	A category (yes/no) for survival.
alone	Categorical	Nominal (or Binary/Dichotomous)	A logical category (True/False) indicating if the person was alone.
survived	Categorical	Nominal (or Binary/Dichotomous)	Represents a group (survived/not survived). The numbers (0 and 1) are just labels for the categories.
class	Categorical	Ordinal	Text representation of pclass (First, Second, Third), showing a meaningful order.
pclass	Categorical	Ordinal	Represents the passenger class (1st, 2nd, 3rd). The classes have a meaningful, ordered ranking.

1) Categorical Columns (Nominal and Ordinal)

These columns represent qualities or attributes and can be divided into groups.

- **Nominal (Names/Labels with no order):**
  - `survived` (0, 1)
  - `sex` (male, female)
  - `embarked` (S, C, Q)
  - `who` (man, woman)
  - `adult_male` (True, False)
  - `deck` (C, NaN)
  - `embark_town` (Southampton, Cherbourg)
  - `alive` (yes, no)
  - `alone` (True, False)
- **Ordinal (Labels with a meaningful order):**
  - `pclass` (1, 2, 3)
  - `class` (First, Second, Third)

2) Discrete Columns

These columns contain numerical data that are countable and can only take on a finite number of values (typically integers).

- `sibsp`
- `parch`

3) Continuous Columns

These columns contain numerical data that can take any value within a range (often includes fractions or decimals).

- `age`
- `fare`

Types of Categorical Features

1. Nominal: No specific order (e.g., colors: Red, Blue, Green).

2. Ordinal: Has an inherent order (e.g., education levels: High School, Bachelor’s, Master’s).

Discrete Features

- Definition: Numerical features that take countable, finite values.

Examples:

- Number of children in a family
- Number of products purchased
- Number of login attempts

Data Features.....

- Which of these are Discrete features?

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
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parch	Discrete	Ratio	Count of parents/children aboard. It must be an integer and the value of 0 is a true zero.
sibsp	Discrete	Ratio	Count of siblings/spouses aboard. It must be an integer and the value of 0 is a true zero.

Mathematical Properties of Discrete Features

Countable values with no intermediate values.

Follows probability mass functions (PMF).

Example: Number of calls received per day:

$$P(X = x) = \frac{\text{Frequency of } x}{\text{Total number of observations}}$$

Continuous Features

- Definition: Numerical features that can take any value within a range.

Examples:

- Height and weight
- Temperature and pressure
- Income and expenses

age	Continuous	Ratio	Can take any value within a given range (e.g., 22.0, 38.0). Zero is an absolute minimum.
fare	Continuous	Ratio	Can take any value within a given range, representing cost.

Mathematical Properties of Continuous Features

Can take infinite values within a range.

Follows probability density functions (PDF).

Example: Height of individuals:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ = Mean
- σ = Standard deviation

Python Implementation

- ‘titanic’ dataset in seaborn

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
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Multiple Choice Questions (MCQs)

Q1: Which feature type is used to represent customer feedback: “Satisfied,” “Neutral,” “Dissatisfied”?

A) Categorical

B) Discrete

C) Continuous

D) Ordinal

Q2: The number of books a person owns is an example of which feature type?

A) Categorical

B) Discrete

C) Continuous

D) Ordinal

Multiple Choice Questions (MCQs)

Q1: Which feature type is used to represent customer feedback: “Satisfied,” “Neutral,” “Dissatisfied”?

A) Categorical ✅

B) Discrete

C) Continuous

D) Ordinal

Q2: The number of books a person owns is an example of which feature type?

A) Categorical

B) Discrete ✅

C) Continuous

D) Ordinal

Fill in the blanks

1. _____ features represent categories or groups.

2. A _____ feature can take any value within a range.

3. The count of items purchased is an example of a _____ feature.

Fill in the blanks

1. _____ features represent categories or groups. (Answer: Categorical)

2. A _____ feature can take any value within a range. (Answer: Continuous)

3. The count of items purchased is an example of a _____ feature. (Answer: Discrete)

Session Summary

In this session, we have discussed about:

- Define categorical, discrete, and continuous features.
- Understand the differences and significance of each type.
- Explore examples with real-world datasets.
- Use Python (Pandas, NumPy, Matplotlib, Seaborn) for data exploration.

Sample E-commerce Order Dataset

Order_ID	Product_Category	Review_Score	Quantity	Price_Per_Unit	Delivery_Region	Discount_Applied	Customer_Age
1001	Electronics	**5**	1	499.99	North	True	34
1002	Clothing	**3**	2	45.50	South	False	22
1003	Books	**4**	5	12.99	West	False	58
1004	Home Goods	**1**	1	89.00	East	True	41
1005	Electronics	**5**	1	129.99	West	False	29
1006	Clothing	**4**	3	25.00	North	True	67
1007	Books	**2**	10	8.50	South	False	19
1008	Home Goods	**5**	1	350.75	East	False	50
1009	Clothing	**3**	2	65.99	North	False	38
1010	Electronics	**4**	1	199.99	West	True	45

Column	Feature Type	Sub-Type	Reasoning
Order_ID	**Categorical**	Nominal	Unique identifier/label. No inherent order or numerical meaning.
Product_Category	**Categorical**	Nominal	Distinct, unordered groups (Electronics, Clothing, etc.).
Review_Score	**Categorical**	**Ordinal**	Scores (1 to 5) have a meaningful, ordered ranking.
Quantity	**Discrete**	Ratio	Countable, whole number of items sold.
Price_Per_Unit	**Continuous**	Ratio	Can take any value within a range (includes decimals like 499.99).
Delivery_Region	**Categorical**	Nominal	Distinct, unordered geographical labels.
Discount_Applied	**Categorical**	Nominal (Binary)	A yes/no or True/False label.
Customer_Age	**Discrete**	Ratio	Countable, whole number of years (often treated as continuous for modeling purposes, but technically a count).

The Iris dataset contains **150 observations** (rows) of Iris flowers, with **three species** observed. For each flower, it records four features related to the physical dimensions of its petals and sepals, plus the species itself.

Iris dataset

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa
4.6	3.4	1.4	0.3	setosa
5.0	3.4	1.5	0.2	setosa
4.4	2.9	1.4	0.2	setosa
4.9	3.1	1.5	0.1	setosa

Column	Example Values	Feature Type	Sub-Type	Reasoning
sepal_length	5.1, 4.9, 7.0	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
sepal_width	3.5, 3.0, 3.2	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
petal_length	1.4, 1.4, 5.1	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
petal_width	0.2, 0.2, 1.8	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
species	Setosa, Versicolor, Virginica	**Categorical**	Nominal	Represents distinct, unordered groups of flowers.

Wine Quality Data

fixed_acidity	volatile_acidity	citric_acid	residual_sugar	chlorides	free_sulfur_dioxide	total_sulfur_dioxide	density	pH	sulphates	alcohol	quality
7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	**5**
7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	**5**
7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	**5**
11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	**6**
7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	**5**
7.4	0.66	0.00	1.8	0.075	13.0	40.0	0.9978	3.51	0.56	9.4	**5**
7.9	0.60	0.06	1.6	0.069	15.0	59.0	0.9964	3.30	0.46	9.4	**5**
7.3	0.65	0.00	1.2	0.065	15.0	21.0	0.9946	3.39	0.47	10.0	**7**
7.8	0.58	0.02	2.0	0.073	9.0	18.0	0.9968	3.36	0.57	9.5	**7**
7.5	0.50	0.36	6.1	0.071	17.0	102.0	0.9978	3.35	0.80	10.5	**5**

Column	Feature Type	Sub-Type	Reasoning
fixed_acidity	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
volatile_acidity	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
citric_acid	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
residual_sugar	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
chlorides	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
free_sulfur_dioxide	**Continuous**	Ratio	Represents concentration (mg/dm^3). It can take any value within a range.
total_sulfur_dioxide	**Continuous**	Ratio	Represents concentration (mg/dm^3). It can take any value within a range.
density	**Continuous**	Ratio	Represents mass per volume (g/cm^3). It can take any value within a range.
pH	**Continuous**	Interval	A measure of acidity/basicity. Zero is arbitrary (it doesn't mean zero acidity).
sulphates	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
alcohol	**Continuous**	Ratio	Represents percentage of alcohol by volume. It can take any value within a range.
quality	**Categorical**	**Ordinal**	Represents a score (typically 0-10) where higher values indicate better quality, establishing a meaningful order.

Breast Cancer Diagnosis Data

mean radius	mean texture	mean perimeter	mean area	mean smoothness	diagnosis
17.99	10.38	122.80	1001.0	0.11840	malignant
20.57	17.77	132.90	1326.0	0.08474	malignant
19.69	21.25	130.00	1203.0	0.10960	malignant
11.42	20.38	77.58	386.1	0.14250	malignant
20.29	14.34	135.10	1297.0	0.10030	malignant
12.45	15.70	82.57	477.1	0.12780	malignant
18.25	19.98	119.60	1040.0	0.09463	malignant
13.71	20.83	90.20	577.9	0.11890	malignant
13.00	21.82	87.50	519.8	0.12730	malignant
12.46	24.04	83.97	475.9	0.11860	malignant

Column	Feature Type	Sub-Type	Reasoning
mean radius	**Continuous**	Ratio	Physical measurement of the cell nucleus, can take any value within a range.
mean texture	**Continuous**	Ratio	Physical measurement (standard deviation of gray-scale values), can take any value within a range.
mean perimeter	**Continuous**	Ratio	Physical measurement, can take any value within a range.
mean area	**Continuous**	Ratio	Physical measurement, can take any value within a range.
mean smoothness	**Continuous**	Ratio	Local variation in radius lengths, a continuous measurement.
diagnosis	**Categorical**	Nominal (Binary)	Represents two distinct, unordered groups: malignant (cancerous) or benign (non-cancerous).

T20 Batting Performance Data

Player	Innings	Runs	Balls Faced	Strike Rate	Hundreds	Dismissal Type	Batting Hand
V Kohli	107	4008	2905	137.90	1	Caught	Right
R Sharma	140	3853	2772	138.60	4	Bowled	Right
B Azam	98	3698	2800	132.07	3	Caught	Right
M Rizwan	85	3135	2400	130.63	1	Run Out	Right
J Buttler	100	2720	1850	147.03	1	LBW	Right
D Warner	99	2894	2000	144.70	1	Caught	Left
C Gayle	79	1899	1220	155.65	2	Stumped	Left
K Williamson	87	2464	2050	120.19	0	Caught	Right
S Narine	92	1100	700	157.14	0	Caught	Left
F du Plessis	50	1675	1250	134.00	0	Bowled	Right

Column	Feature Type	Sub-Type	Reasoning
Player	**Categorical**	Nominal	A unique name/label used to group statistics.
Innings	**Discrete**	Ratio	A count of matches batted. Must be a whole number, and 0 is a true zero.
Runs	**Discrete**	Ratio	A count of runs scored. Must be a whole number, and 0 is a true zero.
Balls Faced	**Discrete**	Ratio	A count of balls faced. Must be a whole number, and 0 is a true zero.
Strike Rate	**Continuous**	Ratio	Calculated as $(Runs / Balls Faced) \times 100$. Can take any value within a range (includes decimals).
Hundreds	**Discrete**	Ratio	A count of innings scoring 100 or more. Must be a whole number.
Dismissal Type	**Categorical**	Nominal	Represents distinct, unordered groups (e.g., Caught, Bowled, LBW, Run Out).
Batting Hand	**Categorical**	Nominal (Binary)	Represents two distinct, unordered groups (Right, Left).

Feature Type	Sub-Type	Definition	Example (Dataset)	Formula (If Applicable)
1. Categorical	**Nominal** (or Binary)	Represents labels, names, or distinct, **unordered** groups. Numbers used are only for labeling (e.g., 0=no, 1=yes).	sex (male/female), diagnosis (malignant/benign), Dismissal Type	N/A (Based on grouping/labeling)
	Ordinal	Represents categories that have a **meaningful order or rank**. The interval between ranks is not necessarily equal or measurable.	pclass (1st, 2nd, 3rd), Review_Score (1 to 5), Quality (0-10 score)	N/A (Based on rank/order)
2. Discrete	**Ratio** (Count)	Data that are **countable** and must be **whole numbers**. The range of possible values is fixed (no values between integers). Zero is a true zero.	Innings, Runs, sibsp (siblings/spouses count)	N/A (Simple count of events)
3. Continuous	**Ratio**	Data that can take **any value** within a given range (decimals allowed). Zero means the complete absence of the property.	sepal_length, fare, alcohol, mean radius	Example: **Strike Rate** $= \frac{\text{Runs}}{\text{Balls Faced}} \times 100$
	Interval	Data that can take **any value** within a given range, but **zero is arbitrary** (it does not mean "nothing").	pH (A measure of acidity, not a count or true zero amount), Temperature (Celsius/Fahrenheit).	N/A (Based on an established scale without a true zero point)

Reference Links

- Python notebook
- How to Create Stunning Data Visualizations in Python: Top 10 Techniques to Learn
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- How Much Statistics Do You Really Need for Data Science?
- Phases of data science and analytics.